

STUDENT PROJECT PROPOSAL

Academic Year 2025 / 2026

1. Project Information

Project Title	ECO Park lot
Program / Department	Engineering
Date of Submission	6/6/2026

Team Members

List all team members with full name, ID, specialization, and role on the team.

#	Full Name	ID	Specialization	Role on Team
1	Ahmed Mostafa Mostafa	240102998	Energy Engineering	Team Leader
2	Mohab Mohamed abdelwahaab	230105667	Electrical and Electronics Engineering	
3	Ahmed Mohamed Mohamed	240102989	Energy Engineering	
4	Mostafa Hisham Ibrahim	240102762	Energy Engineering	
5	Mohamed Safwat Shalapy	230100450	Electrical and Electronics Engineering	
6	Mohamed Zakaria	240103011	Electrical and Electronics Engineering	
7	Esam Ahmed	240103425	Energy Engineering	

Supervisor

Name	Dr. Mariam Ahmed Sameh
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Email	Mariam.sameh@sut.edu.eg
Department	Engineering

Industry Partner / Stakeholder (if any)

If your project is in collaboration with a company, NGO, or government entity, list them here. Leave blank if not applicable.

Organization	
Contact Person	
Nature of Collaboration	

2. Project Overview

Abstract / Summary

150–200 words. Describe the project idea, the problem addressed, and your proposed solution. Avoid technical jargon — write so a non-specialist can understand.

Traditional parking lots consume substantial amounts of electricity from the power grid to support continuous overhead lighting and safety monitoring systems. This heavy reliance leads to high operating costs and avoidable carbon emissions, while the immense natural sunlight hitting the open parking areas and the mechanical movement from passing vehicles are entirely wasted.

To address this challenge, the **ECOPark LOT** project introduces a smart, self-powered hybrid parking infrastructure. The proposed solution features solar panels mounted above parking spaces to harvest clean electricity during the daytime while simultaneously offering protective shade for cars. To guarantee a continuous energy supply when sunlight is unavailable, pressure-sensitive floor tiles are embedded along the entry and exit lanes to generate additional electricity directly from the motion and weight of driving vehicles. A centralized smart control unit processes this dual-source energy and stores it in high-capacity batteries to power the parking lot's automated lighting, occupancy screens, and safety monitors efficiently. Ultimately, this system transitions urban parking facilities from passive energy consumers into active, green energy hubs tailored for future smart cities.

Keywords:

Smart Parking Lot, Hybrid Renewable Energy, Solar Power, Vehicle Movement Energy, Battery Power Storage, Sustainable Infrastructure



3. Problem Statement, Market Need & Industry Relevance

3.1 Problem Definition

100–150 words. Clearly define the problem or gap your project addresses. Be specific — avoid vague phrases like "is challenging."

Traditional parking facilities rely entirely on the centralized electrical grid to power their constant operations, including continuous high-intensity overhead lighting, electronic automated access gates, and safety monitoring systems. This total grid dependency causes substantial, unchecked greenhouse gas emissions and high monthly operating costs for facility operators. Furthermore, conventional configurations suffer from an inherent design inefficiency: they treat parking areas purely as passive infrastructure. Vast expanses of open roof space remain exposed to strong solar radiation without harvesting power, while the significant mechanical energy, pressure, and kinetic vibrations generated daily by heavy vehicles driving through entry and exit lanes are completely lost. There is a distinct absence of a unified engineering framework capable of capturing these multiple, localized environmental energy sources simultaneously and storing them to create an autonomous, self-sustaining parking infrastructure.

3.2 Industrial or Societal Relevance

100–150 words. Explain how this problem affects Egypt's industry, energy, technology, manufacturing, or society at large. Cite a statistic if you can.

This problem directly impacts Egypt's national infrastructure and its strategic environmental mandates. Under Egypt's Integrated Sustainable Energy Strategy (ISES), the nation is legally committed to generating 42% of its total electricity from renewable energy sources by 2035. However, expanding commercial real estate and a domestic construction market expanding rapidly at over 10% annually continue to add heavy, grid-reliant public utilities that threaten these clean energy milestones.

Compounding this, Egypt's commercial sector consumes roughly 8% of the country's total power grid output, with a staggering 43% of that allocation consumed exclusively by domestic and commercial building lighting systems. By leaving urban parking spaces unintegrated into the modern microgrid architecture, municipalities miss an immense opportunity to relieve the national grid. Implementing self-powered parking hubs directly reduces public load, mitigates urban carbon footprints, and secures critical grid stability.



3.3 Market Need / Value Proposition

Who would use or benefit from your solution? What is the size and nature of the potential market or impact?

Market Need & Target Beneficiaries

The primary target beneficiaries who would implement and directly use this hybrid smart solution include:

- **Commercial Shopping Centers & Real Estate Developers:** High-traffic venues (such as the *Mall of Egypt* and *Open-Air Mall Madinaty*) requiring vast parking zones where high continuous foot and vehicle traffic maximize pressure-sensor output.
- **Educational Institutions & Medical Campuses:** Large universities and municipal hospitals operating expansive, open-air parking lots that experience predictable, peak daytime utilization.
- **Smart City Public Infrastructure Projects:** Governmental municipalities designing eco-friendly urban communities that require self-sufficient utility networks.

Market Size & Nature of Impact

The scale of potential impact is immense due to the high density of modern multi-vehicle developments across Egypt's newly established administrative cities. The system replaces costly grid power with local generation. Financially, it yields immediate returns by lowering utility bills and providing vehicle shading that minimizes heat-island effects in open concrete environments. Operationally, it introduces automated space monitoring to guide drivers quickly to open spots, reducing gridlock, idling, and vehicle emissions. Ultimately, the system converts a high-maintenance real estate liability into an active asset class: a green, independent power plant.



4. Objectives & Success Criteria

Each objective must have a matching, measurable success criterion. "Build a system" is not measurable; "Achieve $\geq 85\%$ accuracy on test set" is.

#	Objective	Success Criterion (measurable)
1	Detect available parking spaces automatically.	The system correctly detects the occupancy status of all 6 parking spaces during testing.
2	Automate vehicle entry and exit using a servo-controlled gate.	The gate opens and closes automatically whenever a vehicle enters or exits the parking area.
3	The gate opens and closes automatically whenever a vehicle enters or exits the parking area.	The LCD display shows the correct number of available parking spaces at all times during operation.
4	Reduce energy consumption using automatic lighting control.	The LED lighting system automatically turns ON at night and OFF during daytime using the LDR sensor.
5	Integrate renewable energy sources into the parking system.	The solar panel and piezoelectric system successfully provide energy for the LED lighting system during testing.



5. Innovation & Expected Impact

5.1 What Makes This Project Innovative

80–120 words. What is novel about your approach? Compare briefly with what already exists. Be honest — "first to combine X with Y for Z context" is a valid claim.

The EcoPark Lot introduces a novel approach to urban infrastructure by transitioning traditional parking spaces from energy consumers to self-sustaining power generators. While existing smart parking systems rely purely on external grids to power their IoT sensors and display units, our design is the first to combine piezoelectric energy harvesting from vehicle movement with photovoltaic (PV) solar panels. This dual-source hybrid system guarantees uninterrupted hardware operation. Furthermore, the integration of under-ramp piezoelectric sensors provides a dual function: it dynamically detects vehicle occupancy and weight variations while simultaneously generating micro-power, offering a truly decentralized, eco-friendly solution for modern smart cities.

5.2 Expected Impact

Check all that apply, then describe the most relevant one in 1–2 sentences.

☒	Cost saving / efficiency improvement
☒	Energy or environmental sustainability
☐	Industry collaboration / industrial scale-up potential
☐	Community benefit / social impact
☒	Educational or research contribution
☐	Other (specify below)



Describe your primary expected impact in 1–2 sentences:

The primary impact of EcoPark Lot is the creation of a self-sustaining infrastructure that significantly reduces operational costs and carbon emissions by harvesting renewable energy directly from vehicle interaction. Furthermore, it serves as a practical research model for integrating hybrid energy sources (Piezoelectric and PV) into smart city IoT frameworks, promoting sustainable urban developments.

6. Methodology & Technical Approach

150–250 words plus a diagram. Outline your approach step by step. Mention the major tools, hardware, and software you'll use. A simple block diagram or flowchart is required.

6.1 Approach Overview

The development of the **ECOPARK LOT** prototype follows a structured, step-by-step engineering framework designed to translate hybrid energy harvesting and localized telemetry into a functional physical model

Step-by-Step Implementation Approach

1. **Mechanical Fabrication:** A customized project wooden prototype is constructed to house the parking bays and transit lanes. The 30mm piezoelectric sensors are mounted beneath high-traffic lane thresholds using a series-parallel combined grid optimization (Design B) to maximize micro-ampere current output. Photovoltaic solar panels are built over the bays to capture daylight and provide physical shade.
2. **Power Management Setup:** Electrical paths from the dual-generation sources are isolated. The fluctuating AC voltage spikes from vehicle impacts are flattened via full-bridge rectifiers and custom capacitor storage networks, reducing charging latency from 15 minutes down to 2 minutes. Solar energy passes through a charge controller to keep a central lithium-ion battery bank stable, with down-stream buck regulators distributing smooth 5V and 3.3V direct-current power lines.
3. **Hardware Interfacing:** Proximity tracking is achieved by linking 6 infrared slot modules and 2 gate monitors directly to the digital I/O lines of an **Arduino Mega 2560** controller. Visual



telemetry is routed to a 16x2 LCD screen over a 2-wire synchronous I2C bus network, significantly reducing wire crowding compared to parallel buses.

4. **Software & Optimization:** Firmware is compiled within the Arduino IDE using native C/C++. The code relies entirely on localized edge logic rather than cloud processing to minimize power draw. It utilizes non-blocking boolean tracking flags to manage gate states, protecting the SG90 servo motor barrier from jittering or re-triggering under crawling model vehicles.

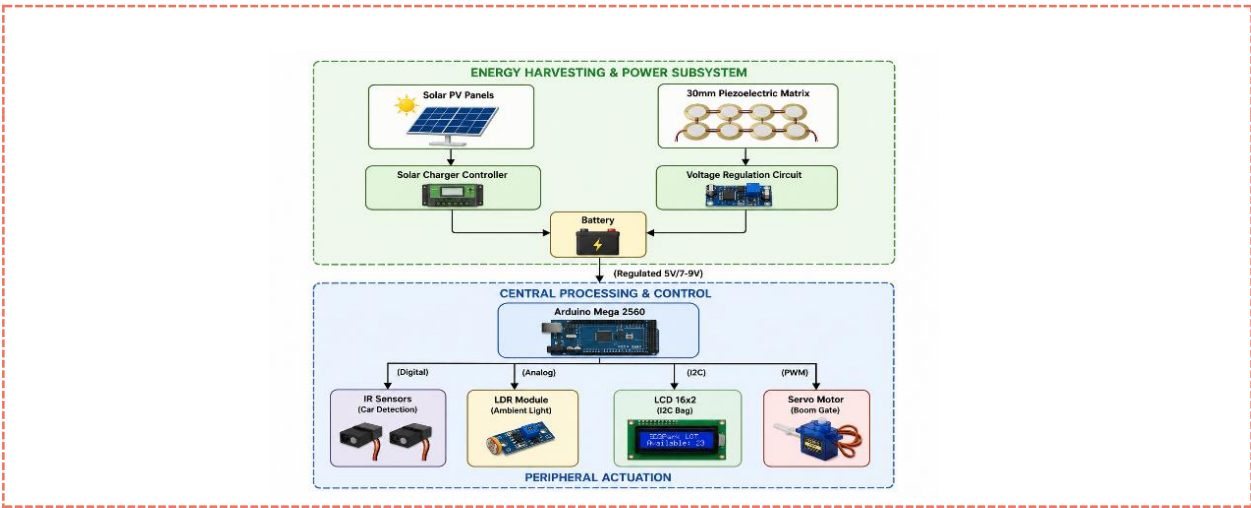
6.2 Tools, Hardware & Software

Adapt to your specialization — examples below are illustrative across disciplines.

Hardware platforms	Arduino
Software & languages	C++ /AutoCAD /Arduino ID
Datasets / standards	public dataset
Methods / algorithms	e.g. PID control / CNN / FEM analysis / IoT protocol ...

6.3 Block Diagram / Process Flow

Insert a system block diagram or process flowchart below.



7. Project Deliverables

Tick all deliverables your project will produce, then add a one-line description of each ticked item.

	Deliverable	Description
✓	Working prototype / hardware system	A smart parking system using Arduino, IR sensors, solar power, and a piezo system
✓	Software / source code	The Arduino C++ source code written to control the gate automatically and display available parking spaces
✓	Technical report & documentation	The complete graduation project book documenting all components and technical details
✓	Final presentation	A PowerPoint presentation summarizing the project idea and results for the final defense
✓	Demonstration video	A short video showing the practical operation of the smart parking .lot and its systems
✓	Dataset / analysis output	Sensor readings and the estimated power generated from the solar panels and piezo system
☐	Other (specify)	No other

8. Timeline & Milestones

Fill the dates and the responsible team member for each phase. Add or remove rows as needed.

Phase / Milestone	Start Date	End Date	Deliverable	Responsible Member
Literature Review & Requirements	20\10\2025	29\10\2025	Study existing smart parking systems, renewable energy integration, and IoT solutions.	



Phase / Milestone	Start Date	End Date	Deliverable	Responsible Member	
Design & Methodology	7\11\2025	19\11\2025	Plan system architecture, select hardware/software, and define technical approach.		
Prototype Development	18\11\2025	25\12\2025	Build the prototype, connect sensors, solar panels, and microcontroller.		
Testing & Validation	25\12\2025	15\1\2026	Conduct functional testing, collect performance data, and refine the system.		
Final Report & Defense	5\1\2026	20\1\2026	Prepare documentation, presentation, and demonstrate the project		
Prototype Development	28\3\2026	16\4\2026	Adding an Arduino with sensors, a first servo motor, and an LCD. And connecting them.		
Testing & Validation	18\4\2026	27\4\2026	Ensuring the systems are interconnected and functioning correctly while resolving problems.		
Final Report & Defense	3\5\2026	5\6\2026	Working on making a book and creating advertisements for the project		



9. Resources & Estimated Budget

9.1 Required Resources

Lab facilities, software licenses, datasets, or any access you'll need from the department.

The ECO Park Lot project requires several hardware and software resources to ensure successful implementation, testing, and operation of the system. The selected components were chosen based on their functionality, availability, reliability, and cost-effectiveness.

Hardware Resources

- Arduino Mega Microcontroller
- LCD 16×2 Display with I2C Module
- IR Sensors
- LDR Sensors
- Piezoelectric Sensor
- Servo Motor (SG90)
- LEDs
- Jumper Wires
- Rechargeable Battery
- Solar Panel
- Charge Controller
- Voltage Regulation Circuit

Software Resources

- Arduino IDE for programming and debugging.
- Personal Computer or Laptop for software development and testing

Laboratory Resources

- Electronics laboratory for hardware assembly and testing.
- Internet access for research and documentation.
- Basic measurement and testing equipment



9.2 Estimated Budget (Bill of Materials)

List every cost item with quantity and price. The total at the bottom must be filled in.

Item / Component	Qty	Unit Cost (EGP)	Total (EGP)
1- Led	30	10/10pieces	30
2- Jumper Wires	20m	10	200
3- IR Infrared Module 3-Pin	9	35	315
4- LDR Module – Optical Sensitive Resistance Light Detection Sensor Module3Pin	2	30	60
5- Servo Motor (SG90)	2	90	180
6- Rechargeable Battery	4	110	440
7- Solar Panel	6	300	1,800
8- Piezoelectric Sensor30mm	30	30	900
9- Solar Charger Controller	1	550	550
10- Voltage Regulation Circuit	1	70	70
11- LCD 16x2 with I2C Module	2	135	270
12- Arduino Mega	1	1,000	1000
13-Project Wooden Prototype	1	1,500	1,500
TOTAL ESTIMATED BUDGET			7,315

Funding Source (if any):



10. Risk Assessment & Safety Considerations

List 3–5 risks that could affect the project. Rate likelihood and impact as Low / Medium / High. Mention any safety precautions on the line below the table.

#	Risk	Likelihood	Impact	Mitigation
1	Communication failure between sensors and Arduino	low	High	Check wiring connections and perform regular testing
2	False vehicle detection due to environmental interference	Mediuem	Meduiem	Adjust sensor sensitivity and proper sensor placement
3	Short circuit in electrical connections	Low	High	Use insulation and protective circuits
4	Overheating of electronic components	Low	Meduiem	Provide proper ventilation

Safety Precautions

Especially relevant for electrical, energy, or hardware projects. List any safety steps (lab protocols, PPE, electrical isolation, fire safety, data protection, etc.).

- proper insulation of all electrical connections.
- Use voltage regulation and battery protection circuits.
- Disconnect power before maintenance.
- Keep components away from heat.
- Regular inspection of sensors, wiring, and battery.
- Secure mounting of the servo motor and gate mechanism.
- Keep flammable materials away from the system.
- Test the system safely before deployment.



11. References

Use IEEE citation style. List at least 5 references — books, journal/conference papers, standards, or reputable websites. Examples below show correct formatting for each source type.

Your References:

- [1] S. B. Prathibha, A. Radhika, and S. S. Shwetha, "IoT Based Smart Parking System Using Renewable Energy Sources," in Proc. Int. Conf. on Inventive Systems and Control (ICISC), Coimbatore, India, 2021, pp. 452–457.
- [2] A. Khaligh, P. Zeng, and C. Wu, "Kinetic energy harvesting using piezoelectric and electromagnetic technologies for smart environments," IEEE Transactions on Industrial Electronics, vol. 57, no. 3, pp. 850–860, Mar. 2010.
- [3] .H. Roshani, S. Dessouky, A. Montoya, and A. T. Papagiannakis, "Energy harvesting from asphalt pavement utilizing piezoelectric technology," Road Materials and Pavement Design, vol. 18, no. 2, pp. 420–431, Apr. 2016
- [4] T. Lin, H. Rivano, and F. Le Mouël, "A Cloud-Based Smart Parking System Based on Internet of Things Technologies," IEEE Access, vol. 5, pp. 24691–24703, Oct. 2017
- [5] US Department of Energy, "Solar Photovoltaic Technology Basics," National Renewable Energy Laboratory (NREL).



12. Supervisor Background, Related Projects & Expertise

To be filled by the supervisor: brief background, area of expertise, and any related projects supervised in recent years. This validates the supervisor-project fit.

12.1 Brief Background & Expertise

12.2 Related Projects Previously Supervised

List 2–3 relevant projects with year and brief description.

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Approval & Signatures

Role	Name	Signature & Date
Supervisor	Mariam Ahmed Sameih	

